

EDUCATION

Ph.D. Student Biochemistry, Molecular Biology and Biophysics Expected 2027
University of Minnesota - Twin Cities
B.S. Biochemistry 2018-2022
University of Minnesota - Twin Cities

RESEARCH EXPERIENCE

Graduate Research with Prof. Burckhard Seelig - UMN, BMBB 2022-present
Characterization of *de novo* primordial-like proteins

- Optimized the expression and purification of >40 *de novo* proteins from libraries of reduced amino acid alphabets in *E. coli*.
- Determined the functional core of a *de novo* variant of interest using protein machine learning and rational design.
- Carried out the biophysical characterization of *de novo* proteins from libraries of reduced amino acid alphabets which act as models for primordial proteins through the use of circular dichroism, X-ray crystallography, and nuclear magnetic resonance.
- Carried out kinetic characterization of >20 *de novo* proteins using ultra-filtration binding assays with radiolabeled adenosine triphosphate.

Structural prediction and analysis of selected *de novo* ligases

- Used protein language models to predict the structures of *de novo* ligases and analyzed the predicted structures to make arguments about their properties and evolutionary trajectories.

Characterization of engineered TIM barrel protein

- Carried out X-ray crystallography on an engineered TIM barrel scaffold protein, obtained and solved a structure.

Undergraduate Research with Prof. Burckhard Seelig - UMN, BMBB 2021-2022

- Studied *de novo* proteins from reduced amino acid alphabet libraries as a way to determine the origins of modern proteins.
- Began the optimization of the purification of >20 *de novo* proteins, carried out gene sequencing, DNA and protein purification, and plasmid transformation.

Undergraduate Research with Prof. Michael Smanski - UMN, BMBB Spring 2020

- Worked towards the vector eradication of pests using novel methods of gene engineering.
- Performed gene sequencing and DNA purification.

TEACHING & VOLUNTEERING EXPERIENCE

Graduate Student Mentor – UMN, BMBB 2022 – present

- Trained and provided guidance for seven undergraduate students in protein expression, purification and characterization. Aided these students in writing honor's theses and applying to graduate schools.

Biochemistry lab teaching assistant – UMN, BMBB 2025

- Taught weekly biochemistry lab courses to undergraduate students in the course BIOC 4025W.

BTI outreach committee volunteer – UMN, BMBB 2022 – 2025

- Co-organized and taught a module in our lab to groups of middle schoolers to demonstrate how liquids with different densities separate and how lipids tend to not mix easily with water.

Bioc 5352 guest lecture – UMN, BMBB 2025

- Served as a guest lecturer for the class Bioc 5352 biotechnology and bioengineering for biochemists, taught the class about how to discover novel enzymes and how to engineer proteins of interest.

Graduate student advisory board – UMN, CBS 2024-2026

- Elected position of student representative for BMBB. Periodically held meetings with the dean of CBS to advocate for students in the college.
- PCE3 seminar planning committee – NASA, PCE3 2026
- Planned monthly seminar series for the NASA research coordination network, Prebiotic Chemistry and Early Earth Environments.

GRANTS, HONORS, AND AWARDS

- 3-minute thesis competition – department-wide winner (BMBB, University of Minnesota) 2023
- 3-minute thesis competition – college-wide winner (CBS, University of Minnesota) 2023
- NASA FINESST grant 2024
- Edith Walter Jones and Robert Jones Fellowship 2024
- Biotechnology Institute Travel Award 2024
- 3-minute thesis competition – department-wide winner (BMBB, University of Minnesota) 2024
- Graduate Student Leadership Award 2025
- Astrobiology Science Conference Travel Grant Award 2026
- Medical School - Council of Graduate Students Career Development Award 2026
- Ross A. Gortner Award 2026

PRESENTATIONS (15)

Oral Presentations

- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: Astrobiology Graduate Conference, “Primordial-like proteins created in a test tube”, San Diego CA, 2023.
- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: Astrobiology Graduate Conference, “Structural insights into the origin of life through model primordial proteins”, Ithaca, NY, 2024.
- **Winslow P.J.**, Erickson S, Seelig B: Biophysical Society Meeting 2026, “Functional characterization of *de novo* primordial-like proteins”, San Francisco, CA, 2026.
- **Winslow P.J.**, Erickson S, Lee KL, Newton MS, Seelig B: Astrobiology Science Conference, “Insights into early protein functionality through primordial model-proteins”, Madison, WI, 2026.
- **Winslow P.J.**, Newton MS, Seelig B: Astrobiology Graduate Conference, “Insights into early protein biochemistry through model primordial proteins”, Tucson, AZ, 2026.
- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: BMBB student seminar, “Emergence of protein function investigated by biophysical characterization of random, *de novo* ATP binders”, Minneapolis, MN, 2023.
- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: Chemical Biology interface training grant flash talk, “Primordial Park: investigating ancient proteins”, Minneapolis, MN, 2023.
- **Winslow P.J.**, Erickson S, Lee KL, Newton MS, Seelig B: 2nd Amano enzyme symposium flash talk, “Characterization of primordial-like proteins from reduced amino acid alphabets”, Minneapolis, MN, 2025.
- **Winslow P.J.**, Newton MS, Seelig B: Chemical Biology interface training grant flash talk, “Structure & function of a primordial model-protein”, Minneapolis, MN, 2026

Poster Presentations

- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: Astrobiology Graduate Conference, “Primordial-like proteins created in a test tube”, San Diego CA, 2023.
- **Winslow P.J.**, Lee KL, Newton MS, Seelig B: Astrobiology Science Conference, “Time travel in a test tube: exploring primordial proteins through structural characterization”, Providence RI, 2024.

- **Winslow P.J.,** Erickson S, Lee KL, Newton MS, Seelig B: 2nd Amano enzyme symposium, “Characterization of primordial-like proteins from reduced amino acid alphabets”, Minneapolis, MN, 2025.
- **Winslow P.J.,** Erickson S, Lee KL, Newton MS, Seelig B: BTI-University of Tokyo joint symposium, “Characterization of primordial-like proteins from reduced amino acid alphabets”, Minneapolis, MN, 2025.
- **Winslow P.J.,** Erickson S, Lee KL, Newton MS, Seelig B: 2025 Central US Synthetic Biology Workshop, “Characterization of primordial-like proteins from reduced amino acid alphabets”, Minneapolis, MN, 2025.
- **Winslow P.J.,** Erickson S, Seelig B: Biophysical Society Meeting 2026, “Functional characterization of *de novo* primordial-like proteins”, San Francisco, CA, 2026.